

Huang Baixiang

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Introduction

Hong Kong University of Science and Technology

Sept 2020 – June 2024

BEng in Computer Engineering

- **Areas of Expertise:** Software Engineering, Machine Learning, Computer Vision, Neural Audio Modeling.
- **Career Objective:** Machine Learning Algorithm Engineer(Computer Vision/Audio), AI-Agent System Engineer

Technologies

Languages: Python, C/C++, SQL, Java, JavaScript/Typescript

Technologies: Pytorch, TensorFlow Lite, Embedded C, ReactJS

Career

Universe Action Limited - AI Engineer

April 2024 - Now

- **Generative Modeling:** Led the R&D of large-scale TTS systems based on **GPT-style Transformer** architectures; implemented **DeepSpeed**, **DDP**, and **AMP** to optimize training efficiency and stability on **4x H200** GPU clusters.
- **Model Adaptation & Expansion:** Spearheaded **LoRA-based** fine-tuning on the GPT backbone to enhance stylistic control; successfully expanded **vocabulary and embedding layers** for new language support, followed by head re-alignment to ensure cross-lingual synthesis quality.
- **Vocoder Optimization:** Optimized pre-trained **GAN/Flow Matching** vocoders to reconstruct high-fidelity audio from GPT **hidden latents**; improved phase consistency and acoustic naturalness through target-domain specific fine-tuning.
- **Deployment & Orchestration:** Migrated inference pipelines from Triton to **Ray Serve**, leveraging **Dynamic Batching** for high-throughput processing and engineering **Streaming Audio Output** to minimize time-to-first-token (TTFT).
- **Agentic Ecosystem:** Developed **Model Context Protocol (MCP)** toolsets to bridge generative audio models with LLM agents, enabling low-latency, end-to-end voice interaction capabilities.

Pivot Biomedical Engineering - AI Engineer

June 2023 - April 2024

- **Medical AI Research:** Developed AI-driven blood pressure measurement systems based on the **Korotkoff sound method**; engineered neural networks to simulate clinical auscultation for precise blood pressure estimation.
- **Spatiotemporal Modeling:** Designed and optimized lightweight temporal architectures, including **LSTM**, **GRU**, and **1D-CNNs**, to capture the dynamic features of arterial pulse sounds during cuff deflation.
- **Edge Optimization & Deployment:** Focused on high-efficiency, lightweight model design for resource-constrained environments; implemented **Post-training Quantization (PTQ)** to convert models into **TensorFlow Lite** format, achieving real-time inference on mobile and embedded devices.
- **Full-stack Integration:** Developed cross-platform mobile applications using **Flutter/Dart** and architected robust backends with **Node.js/Express** and **SQL**, creating an end-to-end prototype from signal acquisition to digital health visualization.

Research

Leveling3D: Leveling Up 3D Reconstruction with Feed-Forward 3D Gaussian Splatting and Geometry-Aware Generation

CUHK Intern

- **Publication:** ArXiv: arxiv.org/abs/2603.16211
- **High-Fidelity 3D Reconstruction:** Developed a hybrid pipeline integrating **2D View-refinement Diffusion** with **3D Gaussian Splatting (3DGS)** to tackle geometric inconsistency in sparse-view scenarios.
- **Systematic Model Optimization:** Conducted exhaustive comparative experiments on **Stable Diffusion** across

multiple fine-tuning regimes, including **LoRA**, **full-parameter tuning**, and the integration of **Adapters/ControlNet** to identify the optimal guidance mechanism for structural consistency.

- **Ablation & Loss Engineering:** Iteratively refined the multi-objective loss function by balancing **MSE**, **LPIPS (Perceptual Loss)**, and **Gram-matrix Loss** through rigorous ablation studies, successfully harmonizing pixel-level accuracy with high-level texture synthesis.
- **Data Curative Pipeline:** Engineered an automated data curation system for large-scale synthetic datasets; developed heuristic filtering and image restoration techniques to eliminate "noisy" generated samples, ensuring training stability and preventing mode collapse.
- **Pipeline Integration:** Validated and integrated the optimized 2D generative priors with feed-forward 3DGS, resulting in a unified framework that significantly enhances both novel view fidelity and 3D geometric coherence.

SAFE-D: A Spatiotemporal Detection Framework for Abnormal Driving Among Parkinson's Disease-like Drivers

CityU Intern

- **Publication:** ArXiv: arxiv.org/abs/2510.17517
- **Multimodal System Development:** Developed a high-fidelity data acquisition platform integrating **CARLA Simulator** with hardware controllers (Logitech G29) to synchronize and capture multimodal driving signals (steering, braking, and acceleration).
- **Cross-Modal Feature Fusion:** Designed an advanced fusion architecture utilizing **Cross-Attention** mechanisms to effectively align and integrate disparate spatiotemporal features from multiple input modalities.
- **Efficiency-Oriented Architecture:** Selected and optimized a **CNN-based backbone** for spatiotemporal feature extraction, balancing high-precision detection (96.8% accuracy) with the low-latency requirements for edge deployment.
- **Clinical Pattern Recognition:** Engineered the pipeline to identify subtle motor symptoms associated with Parkinson's Disease (PD) through time-series analysis of driving behavioral data.

Projects

Cloud VPS Development

Personal websites

- Implemented cloud-based VPS development
- Deploy services with Docker.
- Develop NodeJS and Python Flask Full-stack Program
- Keywords: VPS, Cloud Dev, Docker, NodeJS, Python Flask

Vtorch: Graph-based Visual Interface for Pytorch

Supervisors: Xu Dan, Lionel Parreaux

- Using ReactFlow to generate the flow chart of ML pipeline.
- Design the frontend UI using the ReactJS.
- Design the backend database using NodeJS.
- Keywords: Frontend, Explainable ML, Visualization, TypeScript, ReactJS, NodeJS

Edge-to-Photo Generation with EmbeddedGAN

Supervisors: Chen Qifeng

- Implement the dual-branch architecture Embedded GAN network.
- Develop a new method for noise generation.
- Keywords: Image-to-Image Generation, GAN, Pytorch

Achivement

Honor: Gold Medal at the 2024 HK-Techaton Exhibition